COMPX551 Assignment 4

Muse Headset

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Application and its Purpose

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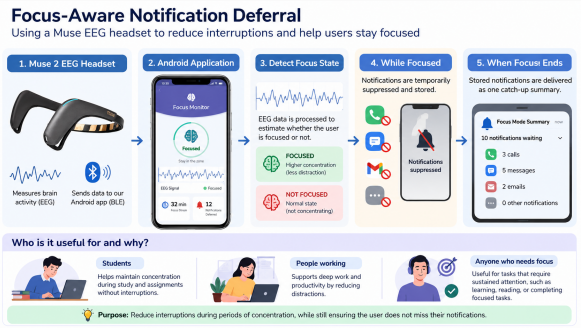
Chosen Device

Muse 2 EEG Headset

Provides EEG data that our Android application uses to estimate the user's focus state.

What does our application do?

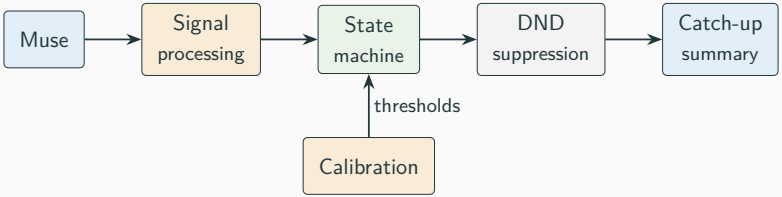
- Estimates whether the user is focused using EEG data.
- Temporarily suppresses incoming notifications while focused.
- Stores notifications during the focused period.



Architecture and device integration

System overview

EEG in, a focus state out, one policy applied consistently downstream.



- Runs entirely on-device, with no data leaves the phone
- One state drives both suppression and the catch up summary
- Fail open at every stage, a bad signal never silently holds a notification

Connecting to the headset

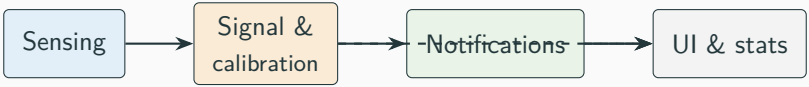
A standalone BLE peripheral, not a paired Wear OS watch, so we use Muse's SDK (`LibMuse`) rather than the Wear OS Data Layer API.



- Registers only the packet types needed, nothing Muse specific leaks out
- Small connection state machine, rescans automatically on a drop
- A dropped or poor contact signal reads as unknown, never silently nothing

Built as four independent modules

The pipeline's own stages became the team's module boundaries — one interface per stage, one owner, developed and tested in isolation.

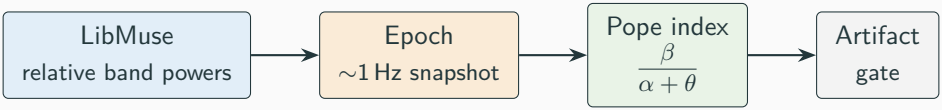


- Every interface has access to a fake demo, so all four people started building day one
- An OSC bridge to an existing app gave real EEG data before BLE worked
- Swapping any module is a one line change, nothing downstream changes

Data processing

From EEG to a focus index

Once a second, the headset's band powers are reduced to a single number.



- Only changing values count, since LibMuse resends the last clean value whenever it is rejecting a window.
- Alpha comes from the temporal electrodes (TP9/TP10), which pick up fewer eye blinks.
- Engagement index from Pope et al. (1995): beta rises and alpha and theta fall as engagement goes up.

Rejecting bad data

Every epoch passes through a gate before it can affect the focus state. Rejected epochs are never used to decide whether to enter or leave a focused state.

Check	Rejects when
Source flags	Headset reports a blink, jaw clench, head movement or muscle noise
Contact	Any of the four electrodes has poor contact
Finite	Index or a band power is NaN or infinite
Muscle	$\beta > 1.3 \times$ the median of the last minute
Hold-off	Within 2 s of an earlier rejection

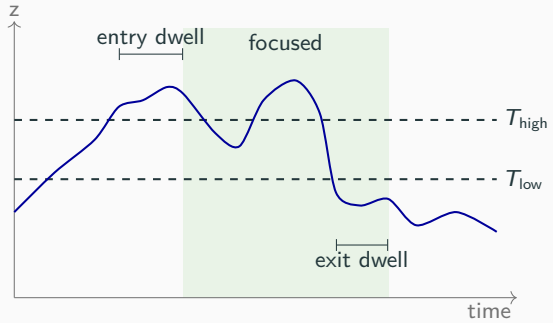
Normalising and deciding

Smoothed, then scored

- Mean of the last 8 accepted epochs
- z-score against the calibration run's mean and SD
- If uncalibrated we use the last 10 minutes instead

Two thresholds, with dwell

- Above T_{high} for 10 s $\Rightarrow$ Focused
- Below T_{low} for 60 s $\Rightarrow$ Not focused
- In between keeps the current state



Calibrating to the user

Fitting the thresholds

- Nine one-minute blocks at low, medium and high load.
- Blocks under 50% task accuracy dropped.
- Thresholds either side of the low–high midpoint.

Quality check

- AUC: chance a high-load epoch beats a low-load one.
- Below 0.6: old thresholds kept.

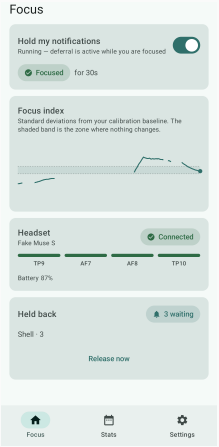
Detecting drift

- Live mean and spread are compared with the calibration run.
- Mean moved > 1 SD, or spread changed $> 2\times$.
- Has to persist for 2 minutes before the user is asked to recalibrate.

Fail open: a poor calibration never starts suppressing notifications.

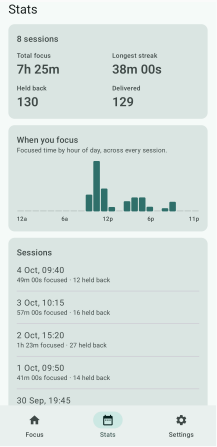
Visualisation and summary data

Live view: what the headset is saying right now



- Focus index: the last two minutes at 1 Hz, in standard deviations from the user's own calibration baseline
- The shaded band is the hysteresis zone between the two thresholds, where nothing changes
- Artifact-rejected epochs *break* the line rather than being interpolated over; the y-range is fixed at ± 3 SD so a calm signal looks calm
- State pill with time-in-state, per-electrode contact quality, and the held-back queue as "app · count", never message content

Summary data: what a week of focus looks like



- One record per session: focused time, longest streak, notifications held back and delivered
- Focused time is measured between state transitions, not by counting FOCUSED epochs, so a dropped packet never shortens the total
- When you focus: focused minutes by hour of day, summed across every session, so the user can see when they do their best work
- Stored on-device as one JSON line per session, written once at session end; every total is computed on read, nothing leaves the phone

Cultural and ethical considerations

Why brainwave data needs care

- **Involuntary:** You cannot choose what your brain gives off.
- **Revealing:** Focus, fatigue and stress can be inferred, and maybe more later.
- **Permanent:** If it is leaked, it cannot be changed.
- Specific legal protections in many jurisdications

What our app keeps

Data	Kept as
Raw EEG	Never stored
Focus decisions	Session totals only
Thresholds	z-scores, not brain power
Notifications	In memory, until released

The app has no internet permission, so it cannot send anything to a server itself.

Consent and privacy

What we do

- Onboarding asks for each permission separately, with a plain reason, and can be skipped
- No session is recorded until the user flicks a switch
- Only the app name and title of each notification are read

Still to consider

- Notifications come from other people who never agreed to the app reading them

Bias in wearables and research

Where bias comes in

- The engagement index comes from a small 1990s lab study, not a diverse population
- Brains differ, for example people with ADHD can show different theta and beta patterns
- Calibration tasks can be biased

How the design responds

- Calibrated per user, so nobody is compared against a population average
- If the index does not work for someone (AUC below 0.6), the app does not act on it
- Poor contact fails open, so notifications are never wrongly held

But failing open still means those users get no benefit.

Data sovereignty and data as taonga

Under Māori data sovereignty, data about people is a taonga: something to be looked after, with rights belonging to the people it comes from.

- The head is **tapu**, the most sacred part of the body, so a device worn on it and data drawn from it deserve extra care.
- **Rangatiratanga:** people control their own data.
- **Kaitiakitanga:** those holding data act as its guardians.

- Keeping data on the user's phone keeps it in their hands and under New Zealand law.
- Cloud backup can send data offshore.
- Research use would need the community involved from the start, not only individual consent.

Reflection and future work

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What we learned

- Fakes behind every interface let four people build in parallel; the real phone still found what no fake could
- Replaying 11 minutes of real reading: 0 % focused with rolling normalisation, 98 % against the calibration baseline with 8-epoch smoothing
- Consumer EEG is noisy: one epoch in three was rejected for blinks, so fail-open everywhere was the right default

Future work

- Evaluation: leave-one-block-out AUC per user, and experience sampling against the state machine in the wild
- Sensor fusion (screen state, foreground app, stillness, calendar) so it degrades gracefully when the headset is off
- Open decisions: a slowly drifting baseline, whether the muscle rule rejects real focus, and consent for the people whose notifications we hold

Thanks